

Call Center Intel

Reference Deck · Idea Title: Call Center Intel — Agentic AI for Sales & Service Conversations
Working prototype, running today on a Medicare tele-sales reference implementation

The Opportunity

Contact centres have every tool to sell and serve — but nothing watches the conversation itself.

Compliance is reviewed weeks after the words were said. QA covers only a sampled handful of calls. New agents ramp on live prospects. Missed calls quietly become lost leads. And AI spend arrives as one untraceable invoice.

Call Center Intel puts a real-time AI layer on every conversation — enforcing compliance in the moment, coaching while the agent is still talking, scoring every call, and attributing the cost of doing so.

Built on an Insurance & Healthcare application, but suited to any contact centre whose agents speak directly to prospects or members.

The Solution — Four Pillars, One Pipeline

One real-time pipeline over the live conversation. The AI listens, assists and measures; the licensed human advises and closes.

1

Agentic Copilot

Works symbiotically with the application, not as a chatbot pushing text back: 23 governed tools that read live business data, drive the screens, draft emails and record updates for one-click approval, and place calls from natural language.

2

Conversation Intelligence

Live transcription with two-layer compliance — deterministic rules in under a second, plus an AI layer that catches paraphrased violations — alongside emotion-shift detection and proactive coaching on every call.

3

Supervisor & Governance

Live monitoring with every alert pinned to the line that triggered it. Business users edit compliance rules, coaching rules and QA scoring criteria themselves — effective on the very next call, with no deployment.

4

Cost Transparency

Per-call and per-agent AI cost from real usage logs, split by feature and drillable to the transcript behind any figure — instead of one opaque vendor invoice at the end of the month.

Technical Architecture

AI & agent layer

LangGraph copilot agent with 23 governed tools • QA evaluator • Claude via OpenRouter for reasoning • Deepgram streaming speech-to-text on both call tracks • live compliance, emotion and coaching analysis over the running transcript

Data & retrieval

Qdrant vector database (5 collections, nomic-embed-text 768-d via Ollama) • hybrid keyword + vector retrieval • MongoDB for application and reference data • hot-reloaded, versioned rule and QA-criteria sets • audit and cost telemetry per AI interaction

Voice, edge & platform

Twilio programmable voice (PSTN inbound & outbound, bidirectional media streams, signed webhooks) • Cloudflare edge with named tunnel • Caddy reverse proxy • Node 20 / Express 5 (TypeScript) API, containerised and cloud-portable

Application & delivery

React 19 / Vite single-page app with Twilio Voice SDK — agent workspace, call panel and copilot • supervisor command centre with live alert feed and mid-call transcript backfill • no-code admin for compliance, coaching and QA scoring criteria • RBAC

How & Where It Will Be Deployed

Environment	Containerised and horizontally scalable — cloud-portable by design. Deployable to any major cloud, hosted independently per client, or on client premises where data residency requires it. The data and AI layers can also run natively on platforms the client already owns, such as Databricks or Snowflake, so the solution sits where their data already lives.
Integration	Every layer is distinct and swappable — this stack is a reference implementation, not a dependency. React could be Angular; MongoDB could be SQL Server or Postgres; the telephony and speech vendors can change without touching the AI layer. REST + SSE surfaces over a governed tool layer front the client's existing CRM, telephony and master data, so it retro-fits onto what they already run.
Security & compliance	No sensitive identifiers requested, by design. Disabled capabilities are withheld from the model rather than discouraged in a prompt. Mandatory automated-call disclosure. Every AI interaction fully audited; PII handling and retention configured per client regulation.
Rollout approach	The four pillars are modular — a client takes what the business needs and leaves the rest. Copilot alone, compliance monitoring alone, or the full platform; each runs without the others. The usual path is to pilot one campaign or queue in shadow mode, switch on the pillars that matter most, then scale across queues and geographies — with the client's own business users tuning rules and QA criteria at every stage.
Ownership & support	EXL Services owns build, deployment, run and support. The client owns the data and the infrastructure it runs on, along with monitoring, latency thresholds and the escalation matrix. Rule and QA-criteria content stays with the client's own business users, who maintain it without a release cycle.

Where We Will Deploy & Sell

Deployment reach — EXL already supports the tele-sales departments of clients across regulated industries; the solution can be deployed into any number of them, and into new pursuits, with no client-specific rebuild

Where EXL already operates

Tele-sales departments of existing clients — the fastest path, because the operation, the data and the compliance obligations are already understood

Health-plan tele-sales and service desks — the domain the prototype already models end to end

Account teams — to identify and confirm the first target department

New logos / pipeline

US health plans — Medicare Advantage & ACA tele-sales desks

Banking & insurance — tele-sales desks operating under conduct rules

Collections & outreach BPOs — contact-practice and disclosure obligations

Retail & telecom — offer and claims compliance, churn-save coaching

Partnerships

GTM Strategy

Positioning

The AI layer for regulated conversations — not a new contact centre, an add-on over the one the client already runs. EXL consults on, builds and runs systems for some of the largest enterprises in the world; this is a complete end-to-end implementation, not a concept deck, so it can be demonstrated as-is from the first meeting of a pursuit.

Sales & channel

Expansion first: open new lines of business with organisations whose core operation is the contact centre, use this as living proof of what AI can do for them, then implement across the rest of their operation. Lead through existing account teams and attach to renewals; the live demo is the primary sales asset.

Pricing

Per-agent-seat subscription for copilot + conversation intelligence • per-minute pricing for automated inbound handling • outcome-linked tier for QA and compliance coverage • bundled into managed-operations pricing on BPaaS deals. Per-call cost telemetry keeps margin predictable.

Launch & marketing

Internal launch at the prototype showcase → reference case study from the first pilot → webinars on real-time compliance for regulated sales → thought leadership on human-in-the-loop AI → demo presence at payer and contact-centre industry events.

WORKING POC • AI CAPABILITY SHOWCASE

Call Center Intel

AI capabilities for customer-facing sales & service operations



Agentic AI Copilot



Conversation Intelligence



Agentic QA

Running today on a reference tele-sales implementation — the domain is swappable: contact centers, financial services, retail, any regulated sales or service operation.



Five problems, and the capability built against each

The tools exist to sell and serve — nothing watches the conversation itself. Every problem below is a cost the operation already pays.



Calls unanswered, hours lost to chasing

Prospects dial when the agent is on another call or off shift. The hours that remain go on callbacks, re-dials and list-working just to find who is still interested.

AI covers inbound — hand-over, ring timeout, nobody online — and the outbound chase: callbacks, re-dials, first-pass qualification. Agents get the hours back for people ready to talk.



Compliance enforced after the fact

By the time anyone reads the transcript, the words have been said and the exposure is on the books. Coaching lags the same way — a week late changes nothing.

Live transcription with in-the-moment compliance flags and coaching against business-edited rules, plus auto-filled records and sentiment — one merged AI pass, not three.



Quality judged from a sample

A supervisor hears a handful of calls per agent, so most issues never surface. The standard lives in each reviewer's head, so scoring drifts and comparison is unreliable.

Automated scorecards on every call against scoring criteria supervisors edit themselves, a live supervision view, and full call records with markers pinned inline.



Ramp-up happens on live prospects

A new agent's first genuinely difficult call is with a real prospect, a real recording and a real compliance obligation. Role-play costs supervisor time and measures nothing.

Practice calls against admin-editable AI personas, graded by the same scorecards as production — so “ready for the phones” becomes a number, not a judgement call.



AI spend with no attribution

Spend arrives as one vendor invoice — no per-call figure, no per-agent figure, no split between features. Cost per call and cost at 100 agents can only be estimated.

Per-call and per-user cost from real usage logs, split by bucket, projected to a 100-agent workforce, and drillable straight to the transcript behind any figure.

Five problems, one platform — the AI gathers, prepares and measures; the licensed human advises and closes.

THE CAPABILITIES

Three pillars, one platform

All three run on the same real-time conversation pipeline — transcript in, actions and insight out.



1

Agentic AI Copilot

An embedded assistant with full tool use: answers from live business data, drives the UI, drafts actions, and places calls — approval-gated, or autonomous in Auto-Pilot mode.



2

Real-Time Conversation Intelligence

Live transcription, two-layer compliance monitoring, sentiment shift detection, and proactive coaching — cost-engineered to run on every call.



3

Supervisor Command Center & AI Governance

Live monitoring with instant mid-call backfill; business users edit the rules, coaching, and QA scoring criteria — no deployment, full audit trail.



Human-in-the-loop by default: AI proposes, humans approve — with an opt-in Auto-Pilot mode where policy allows.

PILLAR 1



Agentic AI Copilot

An assistant that works alongside the agent — sees the live call, uses the tools, drives the UI, and asks permission before it acts.

What it does

- ✓ Embedded conversational assistant with full tool use — queries live business data and answers with rich UI cards, aware of the active call
- ✓ Drafts actions — emails, record updates, notes — for one-click human approval, or executes autonomously in Auto-Pilot mode
- ✓ Voice integration: places and controls phone calls from natural language — “call John Smith”
- ✓ Real-time data capture: entities mentioned in conversation auto-fill forms across screens — including staged records that don't exist yet, created in one click
- ✓ Smart caller intelligence: inbound callers identified and their record opened automatically; ambiguous matches get a disambiguation flow

IN PRACTICE



An agent says “call John Smith.” The copilot finds the record, dials, and the call panel opens as it connects — then drafts the follow-up email for one-click approval.

PILLAR 2



Real-Time Conversation Intelligence

Every second of every conversation, watched by two layers — deterministic rules for speed, AI for nuance.

What it does

- ✓ Live transcription with speaker separation, streamed into the agent's workspace
- ✓ Two-layer compliance: instant deterministic rules (admin-configurable phrases & patterns) — violating sentence highlighted inline with a corrective suggestion in under a second
- ✓ AI review layer catches paraphrased violations keyword rules can't — “you'd be crazy not to take this”
- ✓ Live customer sentiment with shift detection (neutral → confused → positive), surfaced to agent and supervisor
- ✓ Proactive coaching engine: configurable behavioral rules — talk-ratio imbalance, long monologues, missed discovery questions — nudge before the mistake
- ✓ Cost-engineered & resilient: one merged structured LLM call per 15-second window covers sentiment + coaching + compliance; if the LLM drops, rules continue, the UI shows “AI degraded,” and a self-healing probe restores service

IN PRACTICE



An agent paraphrases a guarantee — keyword rules stay silent, but the AI layer flags it and offers a compliant rephrase. If the model drops out, rule-based monitoring never blinks.

PILLAR 3



Supervisor Command Center & AI Governance

The whole operation on one screen — and the AI itself governed by business users, not engineers.

What it does

- ✓ Live monitoring: click any active call and watch the transcript stream in real time — joining mid-call instantly backfills the full conversation
- ✓ Compliance, coaching, and sentiment events pinned to the exact lines that triggered them
- ✓ No-code AI governance: business users edit compliance rules, coaching rules, and the QA scoring criteria (dimensions, weights, disclosure checklists) — effective on the next call, no deployment
- ✓ On-demand AI quality scoring against the custom criteria — evidence-quoted scorecards, strengths, coaching points
- ✓ Full auditability: every AI interaction logged with model, tokens, latency, and output

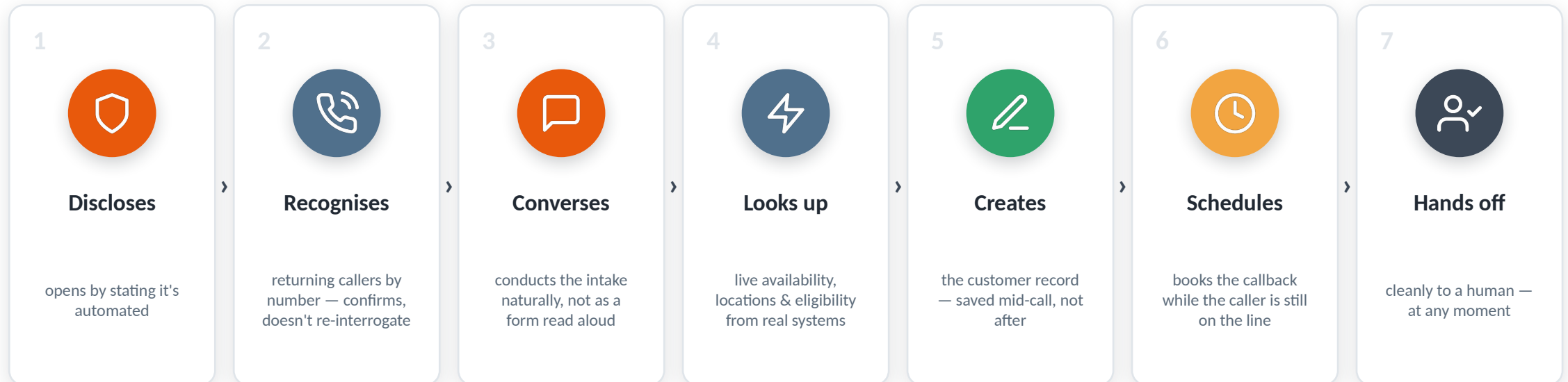
IN PRACTICE



A supervisor joins a call mid-conversation — the full transcript backfills instantly, every alert pinned to its line. Edit a rule, and the very next call obeys it.

What happens on the call

The AI answers when the agent hands it over mid-ring, when the ring times out, or when nobody is online at all. One conversation, seven steps — all of it visible to a supervising human in real time.



The caller experiences a conversation

It ends with a saved customer record, a booked callback, and a full transcript ready for QA.



Engineered for live voice — no dead air

Every system call is time-capped: a slow backend never leaves the caller in silence. Details that surface late are reconciled onto the saved record afterwards.

Grounded in live systems — and the record builds itself

The voice AI calls the same governed tools as the chat copilot, and writes what it learns straight into the customer record.



Real system data, not model knowledge

Availability, locations and eligibility come from live lookups — the model never guesses, and says so when a lookup returns nothing.



Identity injected server-side

The AI acts only as the agent it answers for. Identity is attached by the server, never chosen by the model.



Fields populate live, saved mid-call

The record fills on screen as the caller speaks; mentions are matched to master-data catalogs as structured links, not prose.



Nothing said is discarded

Unmatched detail goes to notes and is flagged for human verification. A returning caller enriches the existing record, never duplicates it.



Voice AI



Governed tool layer



Business systems

By hang-up the conversation is already a clean, structured, deduplicated record — same tools, same guardrails, same audit trail as the chat assistant.

Seamless takeover, guardrails that don't move

AI handles the intake

Human takes over — mid-sentence if needed

one continuous call • one record • one QA analysis

The live transcript streams to the agent's screen while the AI talks — takeover is one click. The customer hears no transfer and repeats nothing.



Always discloses

Every call opens by stating it's automated — non-negotiable, not editable.



Facts, never advice

States what the systems say — no recommendations, no persuasion, no selling.



No sensitive identifiers

Never requested — by design, not by instruction.



Off means off

A disabled capability is withheld from the AI, not discouraged in a prompt.



Deliberately out of scope: outbound AI calling, and AI-made recommendations of any kind. The AI gathers and prepares — the licensed human advises and closes. In a regulated room, that boundary is a feature.

WHY IT MATTERS

What changes in the operation

Three shifts, all of them measurable from the audit and cost telemetry the platform already produces.



Compliance risk down

Violations are caught in the moment, on every call — not in a sampled review weeks later. The rules that catch them are owned by the business.



Agents ramp faster

Real-time coaching and hands-free data capture from day one. New agents get the guidance a supervisor can't give on every call at once.



Supervisors scale

One person effectively present on every live call, with AI they govern themselves — coverage without adding headcount.

“Every conversation coached, supervised, and reviewed — without adding headcount.”

MAP IT TO YOU

What this looks like in your domain

Same three pillars — different rules, vocabulary, and data model. Three examples and an invitation:



Financial services

Suitability & disclosure rules enforced live · KYC details captured mid-call · scoring against your regulatory standards



Retail & telecom

Offer & claims compliance · churn-save coaching nudges · order and account details auto-captured into the CRM



Collections & outreach

Contact-practice rules enforced in real time · required-disclosure checklists · dispute and hardship detection



Your domain

Bring your compliance policies, your coaching playbook, your QA scoring criteria — the three pillars stay the same.

Which pillar would move the needle first in your operation?

ARCHITECTURE

Built from domain-agnostic blocks



Configurable rules engines

Compliance phrases & patterns, coaching behaviors, QA scoring criteria — admin-edited, hot-reloaded, versioned



Structured LLM outputs

Typed, schema-validated responses; merged calls per window — predictable cost and latency



Real-time event streaming

Transcript, alerts, sentiment, coaching pushed live to every screen that needs them



Human-in-the-loop patterns

Draft → approve → execute, with opt-in Auto-Pilot; every action attributable



Tool-using agent framework

23 tools in the reference build — swap the toolset to swap the domain



Audit & cost telemetry

Model, tokens, latency, and output logged per interaction; live stats endpoint

Swapping the vertical means swapping the data model and the rule content — not rebuilding the AI.

INFRASTRUCTURE

Deployment view — every hop, end to end

